

Karan Bajpai

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EDUCATION

B.S. in Computer Science - Pacific University, Forest Grove, OR

August 2020 - May 2024

GPA: 3.7

Awards: Dean's List, Recipient of Future Connect Scholarship

Honors: Magna Cum Laude

General Education - Portland Community College, Portland, OR

September 2018 - June 2020

GPA: 3.90

Awards: President's List, Dean's List, Recipient of Future Connect Scholarship

Relevant Courses: Data Structures / Object Oriented Programming / Operating Systems / Computer Networking / Computer Security / Web Development / Mobile Development / Object Oriented Design / Algorithms / Databases

SKILLS

Languages: Python, JavaScript, Go, SQL, C, C++, Java

Frontend: React, HTML/CSS

Backend: FastAPI, Go

Databases: PostgreSQL, sqlc

Tools: Git, Linux, VS Code

Hardware/Embedded: ROS2, Arduino, Raspberry Pi, OMRON TM Robots, PLC, Ladder Logic

WORK EXPERIENCE

Analog Devices, Inc - Mechatronics Technician II, Equipment Engineering

August 2025 - Present

- Collaborated with lead engineer to design and deploy 4 mobile autonomous robots in a fabrication facility. Each robot has a OMRON LD90 autonomous base + OMRON TM Cobot Arm.
- Programmed and tuned collaborative robots using TMFlow / TMScript during system design, testing, and deployment.
- Executed and modified Python scripts to interface with cobot systems, diagnosing faults and automating control workflows.
- Configured and calibrated robot vision systems for landmark detection, enabling precise localization and automated handling with end effectors.
- Performed remote maintenance using MobilePlanner to edit navigation maps and fine-tune goal placements to optimize fab traversal and tool approach positioning.
- Configured, troubleshooted, and maintained industrial wireless networks, including Phoenix Contact WLAN 1120 radios, IP/Wi-Fi settings, and lab Wi-Fi extenders for robot connectivity.
- Performed electronics assembly, including soldering, wiring, and crimping, following schematics and diagrams.
- Operated autonomous robots in a semiconductor cleanroom environment, using MobilePlanner software to move robots across the fab safely.
- Coordinated with global robotics support teams to diagnose and resolve automation system issues, ensuring uninterrupted production.

PROJECTS

F1 Racing Dashboard (In Development) - React, FastAPI, OpenF1 API

- Building a full-stack Formula 1 dashboard inspired by data-driven sports platforms. This project uses React on the frontend and FastAPI on the backend.
- Planned features include driver comparison tools, and historical season standings.
- Integrated the OpenF1 API to surface race results by session, returning ranked driver standings for any race and season

Collaborative Healthcare Platform - Go, PostgreSQL, sqlc

- Designed the invoices/billing SQL schema for a HIPAA-compliant healthcare platform, implementing soft deletes, audit columns, a custom status ENUM, and foreign keys linking clinics, patients, and appointments.
- Wrote full CRUD SQL query suite with JOINS across patient and clinic tables.
- Defined Go model structs, status constants, and Request/Response/DTO types across service and handler layers.

Chess-playing Robot - C++, Python, Arduino Mega, Raspberry Pi

- Engineered a robotic arm that plays chess by communicating with Stockfish AI API
- Programmed precise motor control using Arduino and system logic in Python on Raspberry Pi
- Showcased multi-device embedded coordination and real-time physical actuation

Shell Emulator - C, Linux System Programming

- Built a UNIX-style shell with support for piping, redirection, and background processes
- Used system calls to handle process creation and inter-process communication
- Reinforced embedded Linux and OS-level control critical for low-resource environments